Rahul Dharmaji

rdh@iika.re · iika.re · linkedin.com/in/rdharmaji · github.com/iikare · US Citizen

Experience

eBay, Inc. / Hardware Systems Software Engineer · San Jose, CA ····· 2024 – present

- Evaluated advanced inference workloads on cutting-edge datacenter GPUs (e.g., Gaudi 3, H100/200, B200/300, MI300/325), delivering detailed performance and latency analyses to support strategic, data-driven procurement decisions.
- Led R&D for applying cutting-edge AI inferencing techniques—such as speculative decoding, disaggregated prefill, and KV cache offloading—to internal LLMs, maximizing production throughput and minimizing TCO.
- Developed LLMBench, an internal, unified benchmarking framework integrating GuideLLM, vLLM, and SGLang, which streamlines performance evaluations by using a one-command pipeline to install dependencies, configure environments, and execute comprehensive test suites.
- Lead author of an in-progress conference paper evaluating the performance of industry-standard inference optimizations on small, cost-efficient, sparse Mixture-of-Experts (MoE) LLMs in production.
- Engineered a one-step Grafana/Prometheus integration for NVIDIA systems, standardizing GPU observability to reduce onboarding friction and enable replicable, shareable benchmarks that accelerate diagnostic cycles.
- Architected an end-to-end ML pipeline to train and deploy a custom Llama 3 LoRA chatbot for internal departmental support, encompassing automated Confluence wiki scraping, LLM-driven Q&A dataset generation, and HuggingFace SFT.

– Languages & Tools · Python, Bash, VLLM, LLMPerf, GuideLLM, Grafana, Prometheus, Kubernetes, Jenkins, Hugging Face, SFT/LoRAs

Embedded & Cyber-Physical Systems Lab / Researcher · Irvine, CA ····· 6/23 – 6/24

- Llm4Plc: Harnessing LLMs for Verifiable Programming of PLCs in Industrial Control Systems ····· (ICSE'24 · 🌐)
  – Architected a state-of-the-art Large Language Model pipeline for Siemens Programmable Logic Controllers, with automated code analysis and formal verification, with a corresponding increase in automated code generation ability by 25%.
- Mic-E-Mouse: Invisible Ears at Your Fingertips: Acoustic Eavesdropping via Mouse Sensors ····· (ACSAC'25 · 🌐)
  – Developed a Machine Learning pipeline to extract audio data from consumer mice using a novel side-channel attack, leading to an 80% speaker classification accuracy, compared to 92% with ground-truth data, as well as a decrease in human-assessed error rate by 83%.
- Llm4Cve: Enabling Iterative Automated Vulnerability Repair with Large Language Models ····· (DSD'25 · 🌐)
  – Architected a state-of-the-art Large Language Model pipeline including fine-tuning with Low-Rank Adaptions for programmatically repairing code vulnerabilities using publicly available CVE data, resulting in a 20% gain in code similarity and an 85% confidence rating when compared to a ground-truth fix.

– Languages & Tools · GPT-3.5/4/4o, Code Llama/Llama 3, Whisper, PyTorch, LoRAs, Python, Bash, C/C++

Vyu Labs, Inc. / Software Engineering Intern · Cupertino, CA ····· 6/21 – 9/21 · 6/22 – 9/22
Installed, tested, and certified development builds for mobile platforms. Ran debugging tools with breakpoints to rectify failing unit tests. Identified bugs and pushed builds onto CI/CD pipelines.

Valkyrie Robotics / Engineering Mentor · Santa Clara, CA ····· 3/18 – 8/23
Provided engineering mentorship to K-12 students towards building functional, adaptable robots.

Education

M.S. Computer Engineering · (GPA: 3.95)

University of California, Irvine ····· 9/23 – 6/24
Algorithms, Data Structures, Computer Architecture, Operating Systems, Convex Optimization, Control & AI, Data Privacy, Deep Learning Compilers, Deep Learning Accelerators

B.S. Computer Engineering · (GPA: 3.90)

University of California, Santa Barbara ····· 9/19 – 6/23

Projects, Skills & Technologies

Interactive Music Visualizer · C++/GLSL ····· 6/20 – present
Built a cross-platform application to artistically visualize live and prerecorded musical input.

Skills & Technologies – C, C++, Python, Rust, LaTeX, Shell, Linux/Unix, Git/GitHub, SQLite, DuckDB, GNU Make, Matlab, ANTLR, PyTorch, TensorFlow, OpenCV, Grafana, Prometheus, Kubernetes, Jenkins and more